

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – DATA CLEANING CHALLENGE

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO3	Assessment:	Tool 2 – Data Cleaning Challenge
Weightage:		Total Marks:	
CO3 Statement: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex. (BL5 and BL6).			
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Section / Batch:	D / 2024	Date of Submission:	09/08/2024
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I BHARANI DHARAN S (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: BHARANI S_

Anomaly Detection (5 Marks)	Cleaning Methods (5 Marks)	Data Transformation (5 Marks)	Analytical Comparison (5 Marks)	Governance Discussion (5 Marks)	Total (25 Marks)

Question 1 – Online Retail Customer Transaction

1. Identify and classify all data quality issues

Issue	Classification	Example
Missing Customer ID	Missing value	NULL for Sneha M
Duplicate transaction	Duplicate	C105 appears twice
Negative quantity	Invalid value	Quantity = -2
Different currencies	Inconsistent data	INR, USD, EUR
Invalid age	Invalid value	Age = 0, 150
Missing age	Missing value	Deepak N
Inconsistent city names	Inconsistent format	Chennai, chennai, CHENNAI
Invalid date	Invalid value	2025-13-01

2. Cleaning strategy for each issue

1. **Missing Customer ID:** Replace with UNKNOWN or generate a new unique ID.
2. **Duplicate transactions:** Remove duplicate records.
3. **Negative quantity:** Remove or verify the transaction.
4. **Different currencies:** Convert all currencies to INR.
5. **Invalid age:** Replace ages below 1 or above 120 with NULL.
6. **Missing age:** Keep as NULL or estimate if reliable information is available.
7. **City names:** Convert all city names to a standard format.
8. **Invalid dates:** Convert invalid dates to NULL and verify the original date.

3. Data standardization and transformation

Examples:

chennai → Chennai

CHENNAI → Chennai

Tiruchirappalli → Tiruchirappalli

Convert all dates to one format:

10-01-2025 → 2025-01-10

Convert all currencies to INR.

Calculate revenue:

Revenue = Quantity × Unit Price

Python Code

```
import pandas as pd
```

```
df = pd.read_csv("retail.csv")
```

```
# Remove duplicates
```

```
df = df.drop_duplicates()
```

```

# Handle missing Customer ID
df["CustomerID"] = df["CustomerID"].fillna("UNKNOWN")

# Remove negative quantity
df = df[df["Quantity"] >= 0]

# Correct invalid age
df.loc[(df["Age"] <= 0) | (df["Age"] > 120), "Age"] = None

# Standardize city names
df["City"] = df["City"].str.strip().str.title()

# Standardize dates
df["PurchaseDate"] = pd.to_datetime(
    df["PurchaseDate"], errors="coerce", dayfirst=True
)

# Currency conversion
rates = {
    "INR": 1,
    "USD": 83,
    "EUR": 90
}

df["UnitPriceINR"] = (
    df["UnitPrice"] * df["Currency"].map(rates)
)

# Calculate revenue
df["Revenue"] = df["Quantity"] * df["UnitPriceINR"]

print(df)
print("Total Revenue =", df["Revenue"].sum())

```

4. Effect of cleaning on total revenue

Before cleaning, the total revenue can be incorrect because:

- Duplicate C105 transaction increases revenue twice.
- C104 has negative quantity.
- USD and EUR cannot be directly added to INR.
- Invalid records may produce incorrect revenue.

After cleaning:

Duplicate records → Removed

Negative quantity → Removed

Currencies → Converted to INR

Valid transactions → Used for revenue

Therefore, **the cleaned total revenue is more accurate and reliable.**

5. Business risks if data is not cleaned

If the data is not cleaned:

1. Revenue may be calculated incorrectly.
2. Top-selling products may be wrongly identified.
3. Customer segments may be incorrectly ranked.
4. Sales reports may contain false information.
5. Management may make incorrect business decisions.

Conclusion

Data cleaning improves the accuracy, consistency and reliability of customer and sales analysis.

Question 4 – Smart City Traffic Monitoring Dataset

1. Assess the quality of the traffic dataset

The dataset contains the following quality issues:

Issue	Classification	Example
Missing timestamp	Missing value	S105
Duplicate reading	Duplicate	S106 appears twice
Negative vehicle count	Invalid value	S103 = -50
Unrealistic speed	Outlier	S104 = 350 km/h
Missing weather	Missing value	S106
Inconsistent road names	Inconsistent data	Anna Salai, anna salai, ANNA SALAI
Invalid Sensor ID	Invalid format	S10A

2. Cleaning techniques for time-series data

1. Remove duplicate sensor readings.
2. Handle missing timestamps.
3. Remove negative vehicle counts.
4. Detect unrealistic speed values.
5. Fill missing weather values as Unknown.
6. Standardize road names.
7. Validate sensor IDs.
8. Convert timestamps into one standard format.

3. Python Code for Data Cleaning

```
import pandas as pd
```

```
# Read dataset
```

```

df = pd.read_csv("traffic.csv")

print("Original Data:")
print(df)

# 1. Remove duplicate readings
df = df.drop_duplicates()

# 2. Remove records with missing timestamp
df = df.dropna(subset=["Timestamp"])

# 3. Remove negative vehicle counts
df = df[df["VehicleCount"] >= 0]

# 4. Remove unrealistic speed
df = df[df["AverageSpeed"] <= 150]

# 5. Fill missing weather
df["WeatherCondition"] = df["WeatherCondition"].fillna("Unknown")

# 6. Standardize road names
df["RoadName"] = df["RoadName"].str.strip().str.title()

# 7. Validate Sensor ID
df = df[df["SensorID"].str.match(r"^S\d{3}$", na=False)]

# 8. Convert timestamp
df["Timestamp"] = pd.to_datetime(
    df["Timestamp"],
    errors="coerce",
    dayfirst=True
)

print("\nCleaned Data:")
print(df)

# Average vehicle count
print("\nAverage Vehicle Count:",
      df["VehicleCount"].mean())

# Average speed
print("Average Speed:",
      df["AverageSpeed"].mean())

```

Main cleaning operations

Duplicate readings → Removed

Missing timestamp → Removed

Negative count → Removed

Speed > 150 km/h → Removed

Missing weather → Unknown

Road names → Standardized

Invalid Sensor ID → Removed

Timestamp → Standardized

4. Effect of cleaning on congestion analysis

Before cleaning, the results may be incorrect because:

- -50 vehicles gives an invalid traffic count.
- 350 km/h gives an unrealistic speed.
- Duplicate S106 readings can increase traffic values.
- Missing timestamps can create gaps in the time series.

After cleaning:

- Vehicle counts become accurate.
- Average speed becomes realistic.
- Congested roads can be identified correctly.
- Traffic signal decisions become more reliable.

Therefore, data cleaning improves the accuracy of congestion analysis.

5. Automated monitoring techniques

The following techniques can prevent future errors:

1. Real-time range checking.
2. Automatic duplicate detection.
3. Sensor health monitoring.
4. Automatic anomaly detection.
5. Alerts for missing data.
6. Validation of Sensor IDs.
7. Automatic timestamp validation.

Conclusion

Cleaning traffic sensor data helps the smart city accurately identify congestion and make better traffic-management decisions.